

**Ahmedabad
University**

CSE623-Machine Learning

Group - 9 ML Code Crafters
Project title: Offline Track
Association Problem [UAV
Videos]

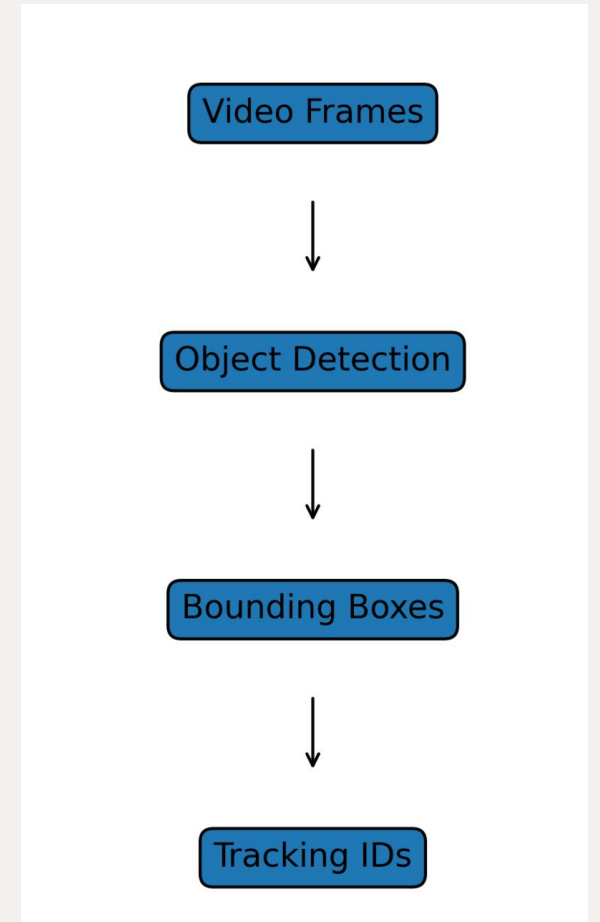
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Problem Statement

- **Objective:** Develop an offline multi-object tracking (MOT) system to handle missing detections and reconstruct object trajectories.
- **The Challenge:** Maintaining continuous trajectories is difficult due to gaps caused by obstructions, sensor malfunctions, and random factors.
- **Impact:** These invasions lead to fragmented tracks, significantly reducing tracking accuracy in UAV surveillance

Dataset

- We used the **VisDrone2019-MOT dataset** for training and evaluation
- Collected using **UAV (drone) cameras** in real-world environments
- Contains:
 - Urban streets, intersections, parking areas
 - Multiple object classes (cars, pedestrians, etc.)
- Provides:
 - Frame-wise image sequences
 - Ground truth annotations for tracking

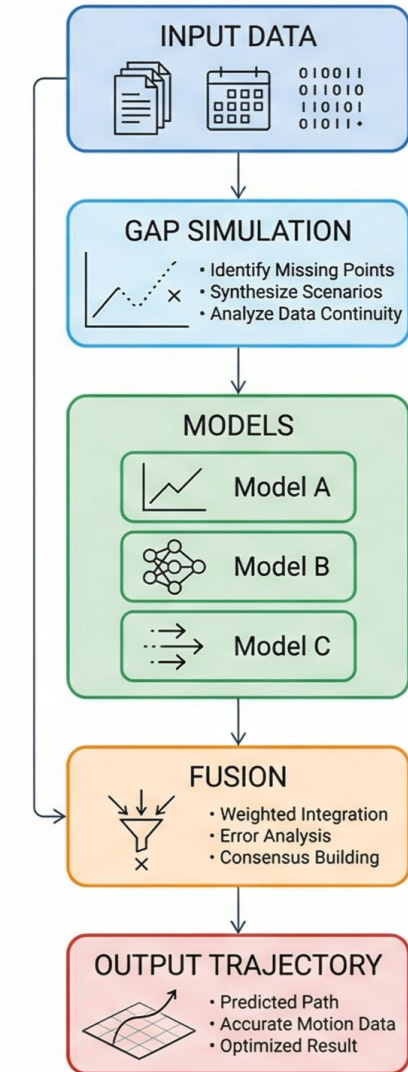


Literature Review

- Existing MOT methods struggle with **handling missing detections effectively**
- Traditional approaches:
 - Tracking-by-detection models (limited in gaps)
 - Kalman Filter (assumes simple motion)
- Research highlights:
 - Hybrid models improve robustness
 - ML-based prediction helps in complex scenarios
- Gap identified:
 - Lack of **adaptive method selection for different gap sizes**

Overview

- Developed an **offline trajectory reconstruction system**
- Uses **VisDrone2019-MOT** dataset
- Combines:
 - Linear Interpolation
 - Kalman Filtering
 - Random Forest Regressor
- Key idea:
 - **Hybrid Decision Fusion** based on gap size



Methodology

Step 1: Data preparation & feature extraction

Step 2: Simulate missing detections

Step 3: Apply predictive models:

- Interpolation (small gaps)
- Kalman Filter (medium gaps)
- ML model (large gaps)

Step 4: Apply smoothing constraints

Step 5: Fuse outputs

Methodology: Core Predictive Models

- Linear Interpolation: Geometry based filling between past and future anchor frames for simple gaps.
- Kalman Filtering (Constant Velocity): State: $[x, y, v_x, v_y, w, h]$.
 - Measurement: $[x, y, w, h]$.
 - Recursively updates when data is present and predicts future states during gaps.
- Machine Learning Regressor: Uses a Random Forest Regressor.
 - Features: Last observed bounding box, historical velocity, and current gap size.

Methodology: Hybrid Decision Fusion

Dynamic Classification: The system automatically selects the best model based on the "seriousness" of the gap:

- **Small Gaps (≤ 2 frames):** Linear Interpolation for high-confidence local recovery.
- **Medium Gaps (3–5 frames):** Kalman Filter combined with Spatio-Temporal smoothing.
- **Big Gaps (> 5 frames):** Weighted Fusion (60% Kalman, 20% Interpolation, 20% Random Forest).

Post-Processing Constraints: Velocity Clamping: Limits sudden spikes in motion.

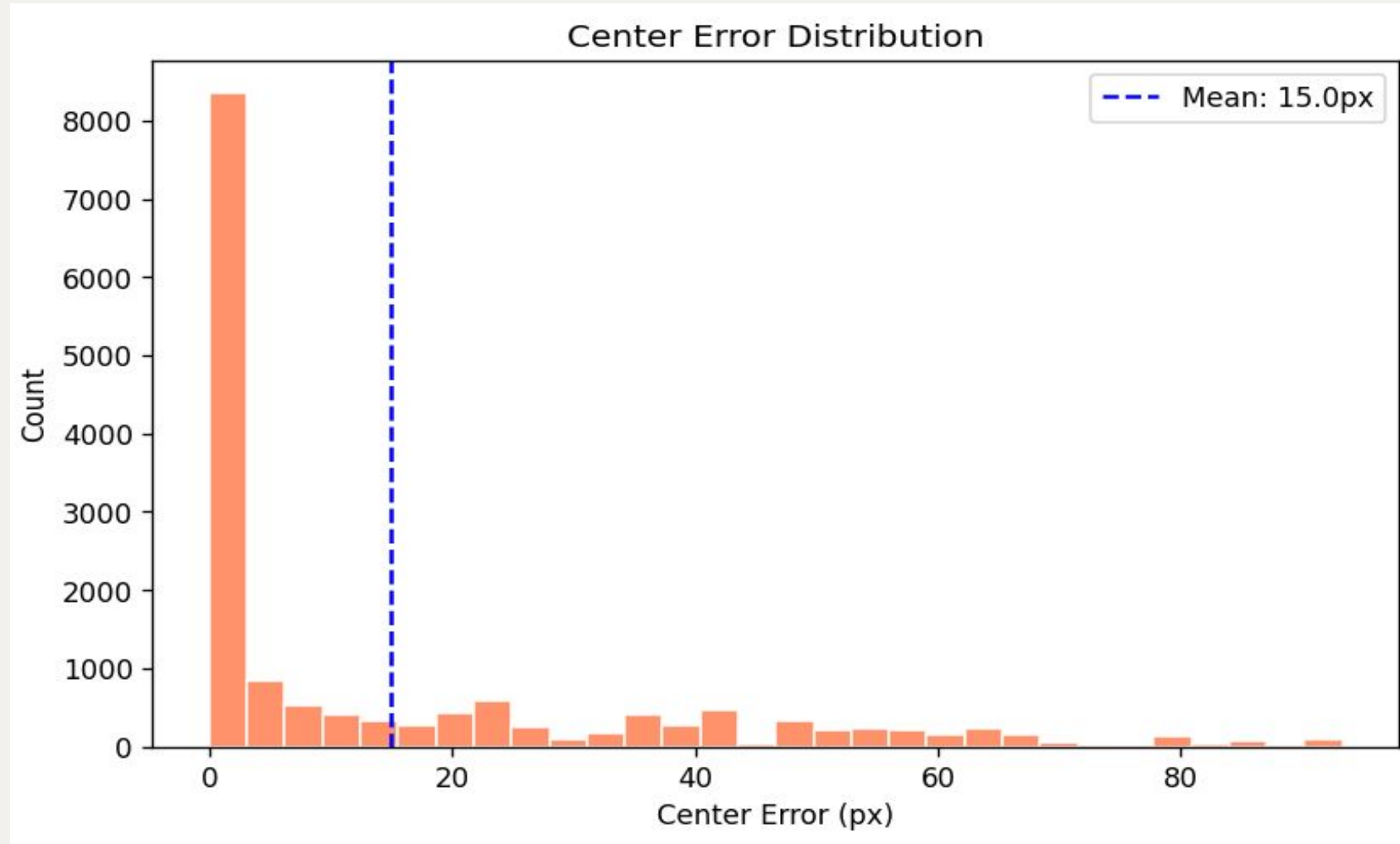
- **Size Consistency:** Prevents bounding boxes from ballooning or collapsing.

Results and Evaluation

Quantitative Metrics Overview:

- **Intersection over Union (IoU):** The system achieved a mean IoU of **0.600**, indicating strong geometric alignment between predicted and actual bounding boxes.
- **Center Error:** The mean Euclidean distance between the predicted center and ground truth center is 15 pixels.
- **Kinematic Realism:** Low "Jerk" (acceleration change rate) values confirm physically realistic movement during trajectory reconstruction.
- **MOTA:** Measures overall tracking performance by combining false positives, false negatives, and identity switches into a single metric
- **IDF1:** Measures accuracy of ID consistency by comparing actual and predicted IDs.

Results and Evaluation





- Green bounding boxes represent those that are already included while Blue ones are the ones predicted from missing detections.

Key Observations:

- accurate tracking
- Good curvilinear motion
- Consistent IDs.

MOTA: 61
IDF1: 58

Future Scope

- **Non-Linear Dynamics:** Implementing Extended or Unscented Kalman Filters to better model complex, sharp drone maneuvers.
- **Deep Learning Expansion:** Moving from Random Forest to architectures like LSTMs or Transformers for better long-term temporal dependencies.
- **Real-time Deployment:** Optimizing the fusion engine for live UAV surveillance rather than strictly offline post-processing.
- **Multi-Sensor Fusion:** Integrating thermal or infrared data to assist in tracking during low-visibility conditions.

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THANK YOU